

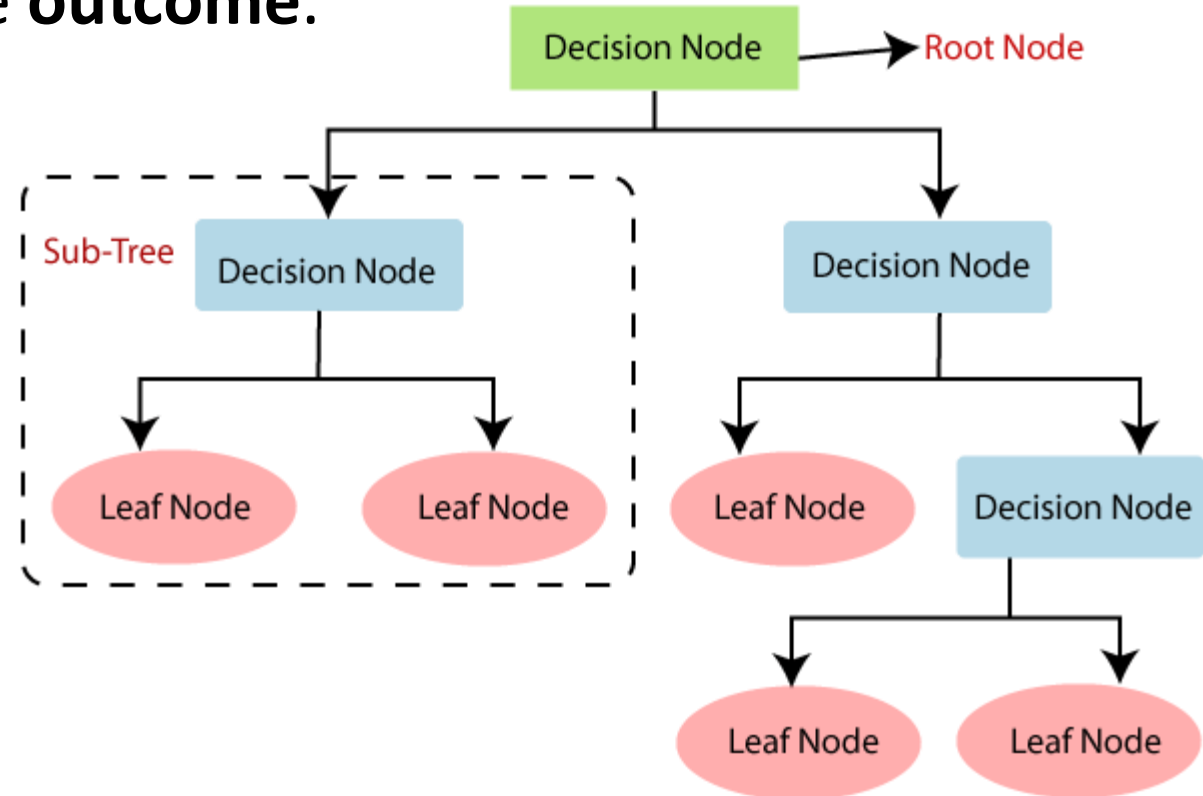
Decision tree

Decision Trees

- Decision tree is a simple but powerful learning Paradigm.
- It is a type of classification algorithm for supervised learning.
- A decision tree is a tree in which each branch node represents a choice between a number of alternatives and each leaf node represents a decision.
- A node with outgoing edges is called an **internal** or **test node**. All other nodes are called **leaves** (also known as **terminal** or **decision nodes**).

Decision tree classification

- internal nodes represent the **features** of a dataset,
- branches represent the **decision rules** and
- each leaf node represents the **outcome**.



Decision Tree Classification Algorithm

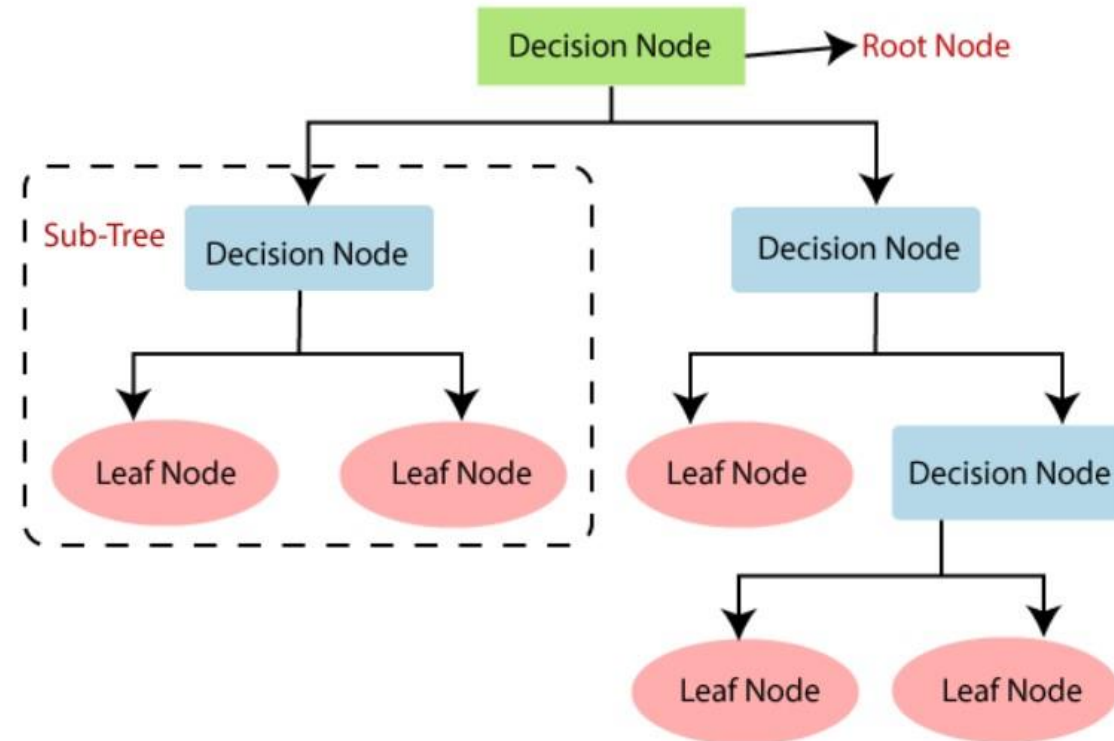
- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but mostly it is preferred for solving **Classification problems**.
- In a Decision tree, there are two nodes, which are the Decision Node and Leaf Node. Decision nodes are used to make any decision and have multiple branches, whereas Leaf nodes are the output of those decisions and do not contain any further branches.
- The decisions or the test are performed on the basis of features of the given dataset.
- It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.
- It is called a decision tree because, similar to a tree, it starts with the root node, which expands on further branches and constructs a tree-like structure.
- In order to build a tree, we use the **ID3, Random Forest** or (Classification And Regression Tree) **CART algorithm**.

Why use Decision Trees?

- There are various algorithms in Machine learning, so choosing the best algorithm for the given dataset and problem is the main point to remember while creating a machine learning model. Below are the two reasons for using the Decision tree:
- Decision Trees usually mimic human thinking ability while making a decision, so it is easy to understand.
- The logic behind the decision tree can be easily understood because it shows a tree-like structure.

Decision Tree Terminologies

- **Root Node:** Root node is from where the decision tree starts. It represents the entire dataset, which further gets divided into two or more homogeneous sets.
- **Leaf Node:** Leaf nodes are the final output node, and the tree cannot be segregated further after getting a leaf node.
- **Splitting:** Splitting is the process of dividing the decision node/root node into sub-nodes according to the given conditions.
- **Branch/Sub Tree:** A tree formed by splitting the tree.
- **Pruning:** Pruning is the process of removing the unwanted branches from the tree.
- **Parent/Child node:** The root node of the tree is called the parent node, and other nodes are called the child nodes.



Construction of decision tree

- 1) First test all attributes and select the one that will function as the best root.
- 2) Break-up the training set into subsets based on the branches of the root node.
- 3) Test the remaining attributes to see which ones fit best underneath the branches of the root node.
- 4) Continue this process for all other branches until get the complete decision tree.

Decision tree

- **Step 1:** Begin the tree with the root node, says S, which contains the complete dataset.
- **Step 2:** Find the best attribute in the dataset using **Attribute Selection Measure (ASM)**.
- **Step 3:** Divide the S into subsets that contains possible values for the best attributes.
- **Step 4:** Generate the decision tree node, which contains the best attribute.
- **Step 5:** Recursively make new decision trees using the subsets of the dataset created in step -3. Continue this process until a stage is reached that further classifying the nodes is not possible and called the final node as a leaf node.

Our Problem :

Name	Gender	Car Ownership	Travel Cost	Income Level	Transportation
Abhi	Male	1	Standard	High	?
Pavi	Male	0	Cheap	Medium	?
Ammu	Female	1	Cheap	High	?

Training Set

Attributes				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

Calculate the entropy of the total dataset

- First compute the Entropy of given training set.

Probability:

Bus : $4/10 = 0.4$

Train: $3/10 = 0.3$

Car: $3/10 = 0.3$

$$E(S) = -P(I) \log_2 P(I)$$

$$E(S) = -(0.4) \log_2 (0.4) - (0.3) \log_2 (0.3) - (0.3) \log_2 (0.3) = 1.571$$

Attributes				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car



Attributes	Classes
Gender	Transportation
Male	Bus
Male	Bus
Female	Train
Female	Bus
Male	Bus
Male	Train
Female	Train
Female	Car
Male	Car
Female	Car

Split the dataset on 'Gender' attribute

Attributes	Classes
Gender	Transportation
Male	Bus
Male	Bus
Male	Bus
Male	Train
Male	Car

Probability:

Bus : $3/5 = 0.6$

Train: $1/5 = 0.2$

Car: $1/5 = 0.2$

Attributes	Classes
Gender	Transportation
Female	Train
Female	Bus
Female	Train
Female	Car
Female	Car

Probability:

Bus : $1/5 = 0.2$

Train: $2/5 = 0.4$

Car: $2/5 = 0.4$

$$Gain(S,A) = E(S) - I(S,A)$$

$$I(S,A) = 1.522 * (5/10) + 1.371 * (5/10)$$

$$Gain(S,A) = 1.571 - 1.447$$

$$= 0.12$$

$$E(S_v) = -0.6 \log_2 (0.6) - 0.2 \log_2 (0.2) - 0.2 \log_2 (0.2)$$

$$= 1.522$$

$$E(S_v) = -0.2 \log_2 (0.2) - 0.4 \log_2 (0.4) - 0.4 \log_2 (0.4)$$

$$= 1.371$$

Attributes				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car



Attributes	Classes
Car Ownership	Transportation
0	Bus
1	Bus
1	Train
0	Bus
1	Bus
0	Train
1	Train
1	Car
2	Car
2	Car

Split the dataset on 'ownership' attribute

Attributes	Classes
Car Ownership	Transportation
0	Bus
0	Bus
0	Train

Probability
Bus : $2/3 = 0.6$
Train: $1/3 = 0.3$
Car: $0/3 = 0$
Entropy = 0.918

Attributes	Classes
Car Ownership	Transportation
1	Bus
1	Train
1	Bus
1	Train
1	Car

Probability
Bus : $2/5 = 0.4$
Train: $2/5 = 0.4$
Car: $1/5 = 0.2$
Entropy = 1.522

Attributes	Classes
Car Ownership	Transportation
2	Car
2	Car

Probability
Bus : $0/2 = 0$
Train: $0/2 = 0$
Car: $2/2 = 1$
Entropy = 0

$$\begin{aligned} \text{Gain}(S,A) &= E(S) - I(S,A) \\ I(S,A) &= 0.918 * (3/10) + \\ & 1.522 * (5/10) + 0 * (2/10) \end{aligned}$$

$$\begin{aligned} \text{Gain}(S,A) &= 1.571 - 1.0364 \\ &= \mathbf{0.534} \end{aligned}$$

- If we choose **Travel Cost** as splitting attribute,

$$\text{-Entropy for Cheap} = 0.722$$

$$\text{Standard} = 0$$

$$\text{Expensive} = 0$$

$$\text{IG} = 1.21$$

- If we choose **Income Level** as splitting attribute,

$$\text{-Entropy for Low} = 0$$

$$\text{Medium} = 1.459$$

$$\text{High} = 0$$

$$\text{IG} = 0.695$$

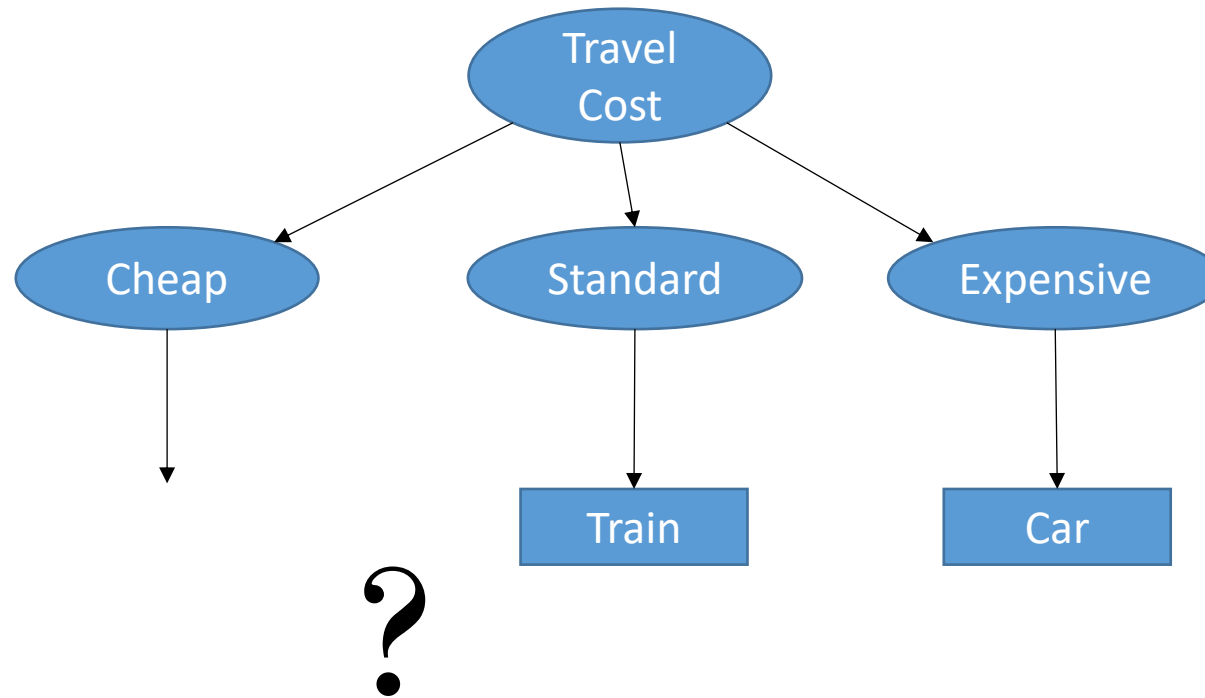
Attribute	Information Gain
Gender	0.125
Car Ownership	0.534
Travel Cost	1.21
Income Level	0.695

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Cheap	Male	0	Low	Bus
Cheap	Male	1	Medium	Bus
Cheap	Female	1	Medium	Train
Cheap	Female	0	Low	Bus
Cheap	Male	1	Medium	Bus

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Standard	Male	0	Medium	Train
Standard	Female	1	Medium	Train

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Expensive	Female	1	High	Car
Expensive	Male	2	Medium	Car
Expensive	Female	2	High	Car

Diagram : Decision Tree



Iteration on Subset of Training Set

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Cheap	Male	0	Low	Bus
Cheap	Male	1	Medium	Bus
Cheap	Female	1	Medium	Train
Cheap	Female	0	Low	Bus
Cheap	Male	1	Medium	Bus

Probability:

Bus : $4/5 = 0.8$

Train: $1/5 = 0.2$

Car: $0/5 = 0$

$$E(S) = -P(I) \log_2 P(I)$$

$$E(S) = -(0.8) \log_2 (0.8) - (0.2) \log_2 (0.2) = 0.722$$

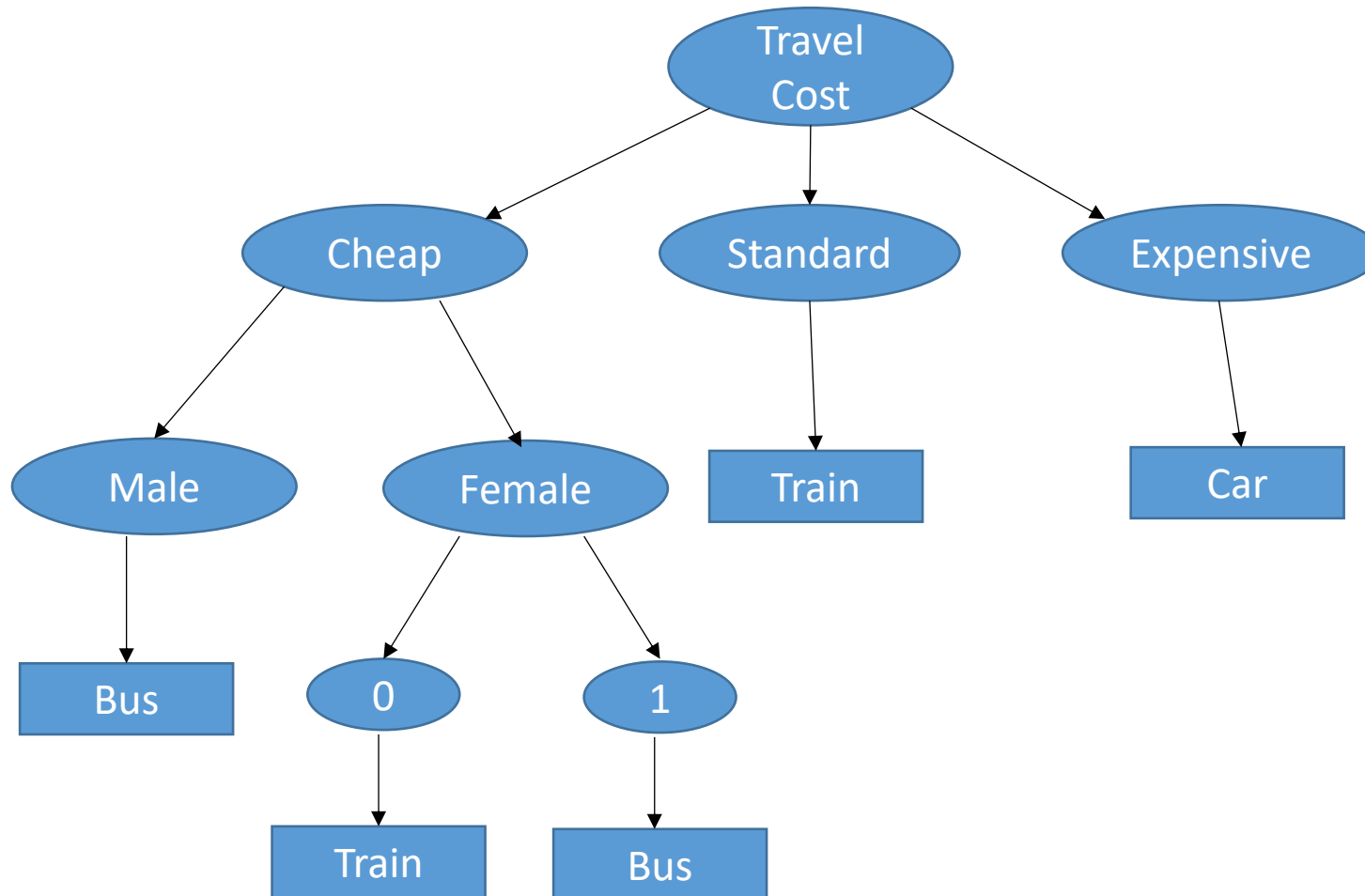
- If we choose **Gender** as splitting attribute,
 - Entropy for Male = 0
 - Female = 1
 - IG = 0.322**
- If we choose **Car Ownership** as splitting attribute,
 - Entropy for 0 = 0
 - 1 = 0.918
 - IG = 0.171**
- If we choose **Income Level** as splitting attribute,
 - Entropy for Low = 0
 - Medium = 0.918
 - IG = 0.171**

Attributes	Information Gain
Gender	0.322
Car Ownership	0.171
Income Level	0.171

Attributes			Classes
Gender	Car Ownership	Income Level	Transportation
Male	0	Low	Bus
Male	1	Medium	Bus
Male	1	Medium	Bus

Attributes			Classes
Gender	Car Ownership	Income Level	Transportation
Female	1	Medium	Train
Female	0	Low	Bus

Diagram : Decision Tree



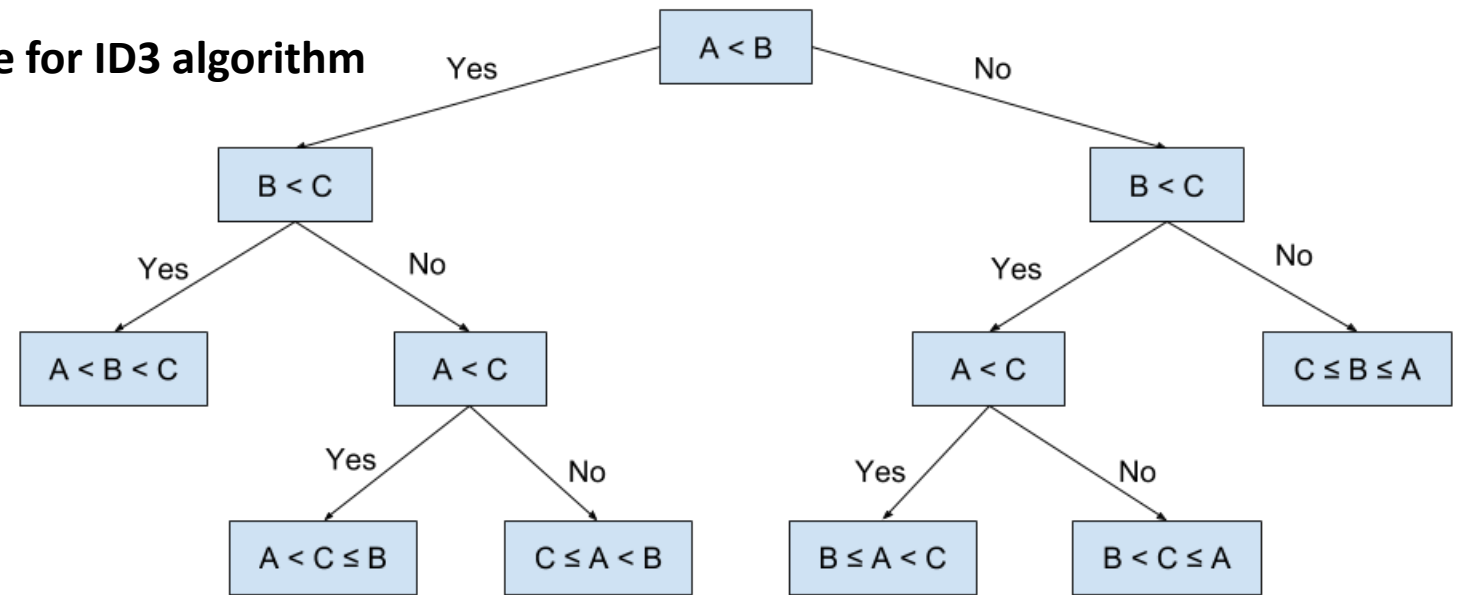
Solution to Our Problem :

Name	Gender	Car Ownership	Travel Cost	Income Level	Transportation
Abhi	Male	1	Standard	High	Train
Pavi	Male	0	Cheap	Medium	Bus
Ammu	Female	1	Cheap	High	Bus

Drawbacks of ID3

- Data may be over-fitted or over-classified, if a small sample is tested.
- Only one attribute at a time is tested for making a decision.
- Classifying continuous data may be computationally expensive.
- Does not handle numeric attributes and missing values.

Example for ID3 algorithm



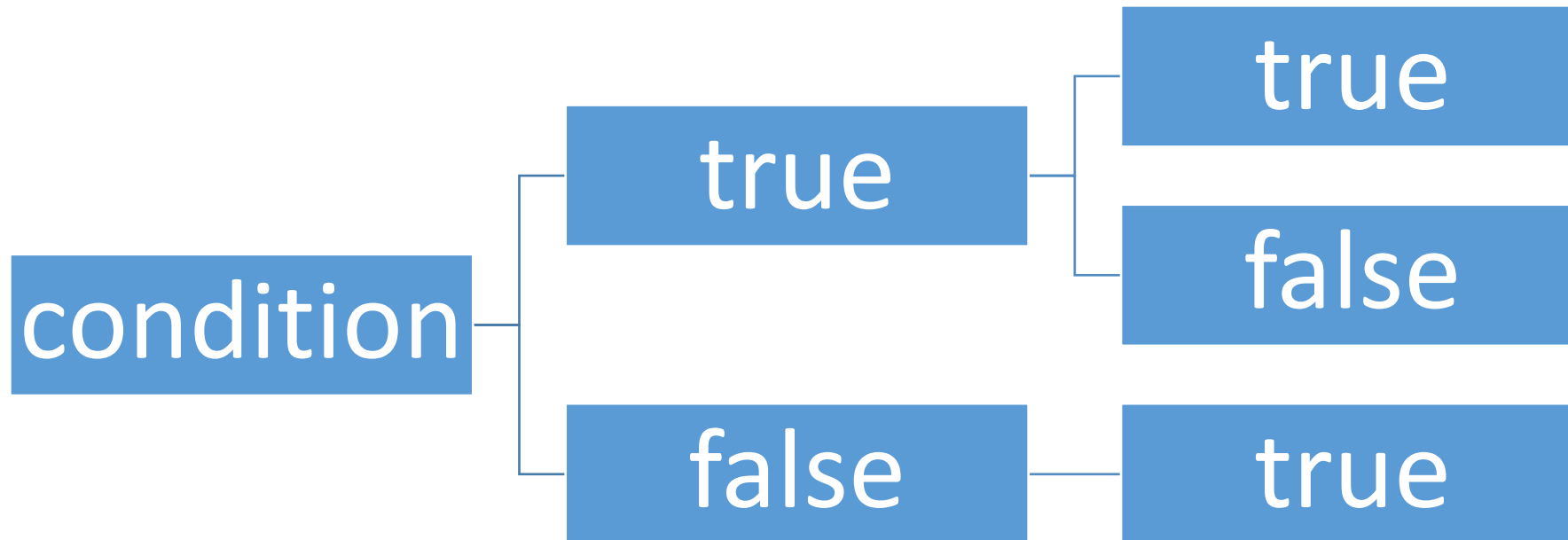
Comparison between ID3 and CART:

Algorithms	ID3	CART
Type of data	Categorical	continuous and nominal attributes data
Speed	Low	Average
Pruning	No	Post pruning
Boosting	Not supported	Supported
Missing Values	Can't deal with	Can deal with
Formula	Use information entropy and information Gain	Use Gini diversity index

CART (Classification and Regression Tree algorithm)

- A decision tree simply asks a question, and based on the answer (Yes/No), it further split the tree into subtrees.
- mimic human thinking ability
- easy to understand

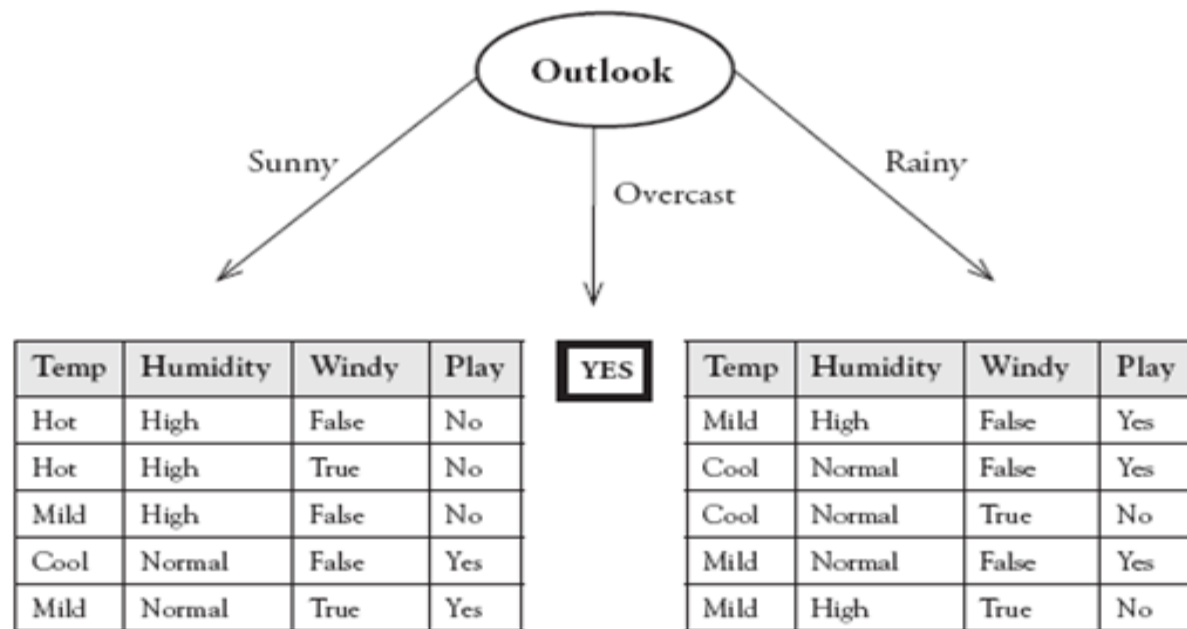
$$\text{Gini_Index}(T) = 1 - \sum_{i=1}^m P_i^2$$



Decision tree

Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	Normal	True	??

Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No



Temp	Humidity	Windy	Play
Hot	High	False	No
Hot	High	True	No
Mild	High	False	No
Cool	Normal	False	Yes
Mild	Normal	True	Yes

Attribute	Rules	Error	Total Error
Temp	Hot → No	0/2	1/5
	Mild → No	1/2	
	Cool → Yes	0/1	
Humidity	High → No	0/3	0/5
	Normal → Yes	0/2	
Windy	False → No	1/3	2/5
	True → Yes	1/2	

Temp	Humidity	Windy	Play
Mild	High	False	Yes
Cool	Normal	False	Yes
Cool	Normal	True	No
Mild	Normal	False	Yes
Mild	High	True	No

Overcast	4	Yes	4
		No	0
Sunny	5	Yes	2
		No	3
Rainy	5	Yes	3
		No	2

High	7	Yes	3
		No	4
Normal	7	Yes	6
		No	1

Attribute	Rules	Error	Total Error
Humidity	High → No	3/7	4/14
	Normal → Yes	1/7	

Attribute	Rules	Error	Total Error
Outlook	Sunny → No	2/5	4/14
	Overcast → Yes	0/4	
	Rainy → Yes	2/5	

Attribute	Rules	Error	Total Error
Temp	Hot → No	2/4	5/14
	Mild → Yes	2/6	
	Cool → Yes	1/4	

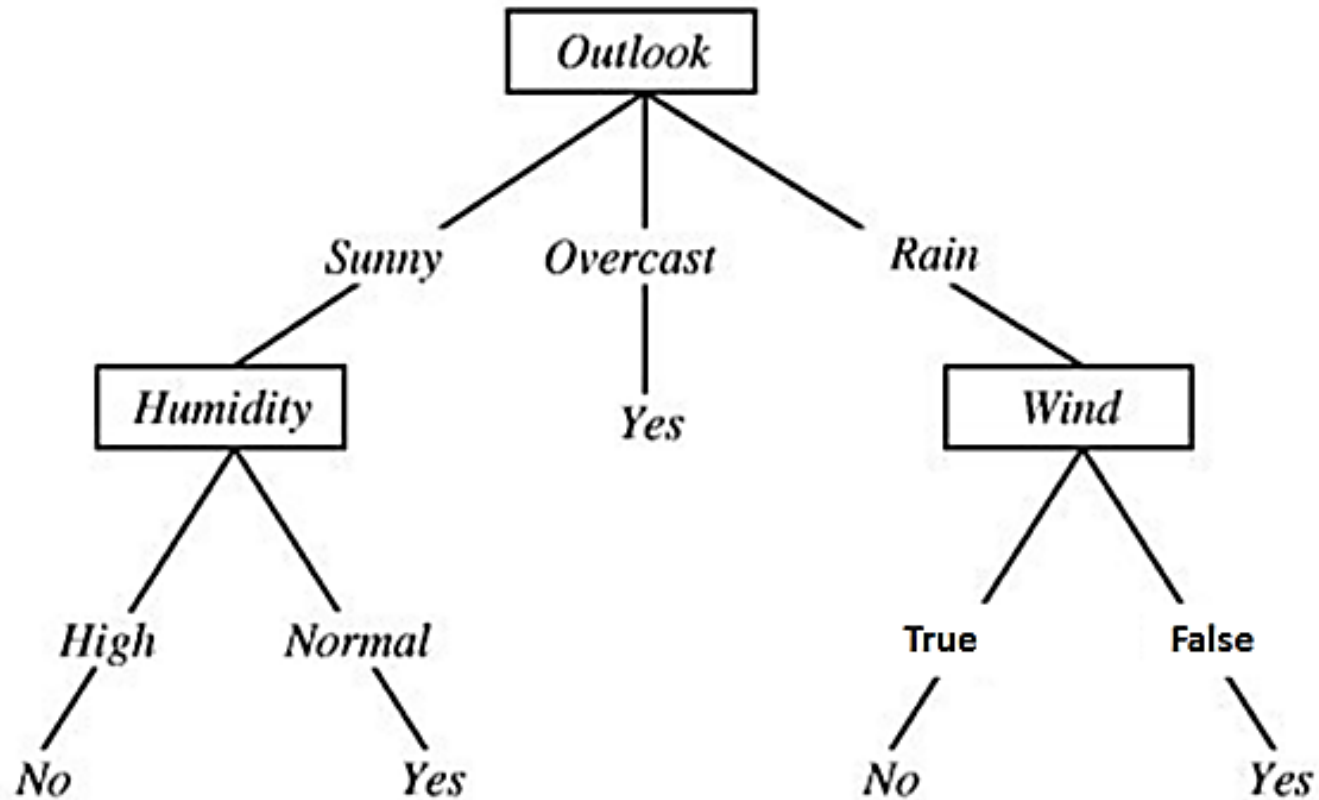
Attribute	Rules	Error	Total Error
Outlook	Sunny → No	2/5	4/14
	Overcast → Yes	0/4	
	Rainy → Yes	2/5	
Temp	hot → No	2/4	5/14
	Mild → Yes	2/6	
	Cool → Yes	1/4	
Humidity	high → No	3/7	4/14
	Normal → Yes	1/7	
Windy	False → Yes	2/8	5/14
	True → No	3/6	

Windy			
False	8	Yes	6
		No	2
True	6	Yes	3
		No	3

Attribute	Rules	Error	Total Error
Windy	True → No	3/6	5/14
	False → Yes	2/8	

- The attributes **Outlook** and **Humidity** have the same minimum error that is 4/14.
- The outlook attribute has one rule which generates zero error that is the rule **Overcast → Yes**.
- Hence, we consider the Outlook as the splitting attribute.
- Humidity has the lowest error
- Windy has the lowest error.

Decision tree



Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	Normal	True	Yes

Thank You!